

Channel Carving: Low-Resistance Attractor Formation Unifying Steering, Guidance, and Collapse

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Abstract

Several behaviors of large models that are usually studied apart — **steerability** (how easily a model can be pushed to a non-default behavior), **guidance** (how strongly inference-time conditioning shifts outputs), **mode collapse** (loss of output diversity under repeated self-training), and **representation consolidation** (why a learned behavior eventually “sticks”) — share one dynamics. We formalize it as **Channel Carving**, the AAMT *Cardboard Channels* theory (Core Premise §4) rendered as a model of attractor basins in the output/representation manifold. Default behaviors are deep, low-resistance **channels** carved by the prior (pre-training distribution, collective data); a desired non-default behavior is a new channel that must be carved against an initial resistance and then consolidates with time constant τ as $R_{\text{new}}(t) =$

$R_0 \cdot e^{-t/\tau}$. We show that (i) inference-time **steerability** equals the flow ratio $\Phi = \text{intention_force} / \text{channel_resistance}$ (guidance strength over prior strength), recovering classifier-free guidance and activation steering as special cases; (ii) **mode collapse** is the deepening of default channels when training flow re-enters them (the same $e^{-t/\tau}$ law, run on the wrong channel), tying this paper to the anti-collapse guard of WP-04; and (iii) **consolidation/permanence** is channel-depth saturation, recovering the **4 τ permanence** rule (Foundations Paper II; 64-State Thm 5.1) as the time for a new channel to become low-resistance. The single τ -law thus governs steering, guidance, collapse, and permanence — previously four separate accounts.

Keywords: attractor dynamics, steerability, classifier-free guidance, mode collapse, model collapse, consolidation, representation engineering, mechanistic interpretability, hysteresis

1. Introduction

Ask four sub-fields why a model does or doesn't change its behavior and you get four answers. Steering/representation-engineering says: add a vector. Guidance says: scale the conditioning term. Model-collapse says: self-training narrows the distribution. Continual-learning says: representations decay unless consolidated. We argue these are one phenomenon viewed from four sides, and the AAMT *Cardboard Channels* theory (Core Premise §4) already names it.

The metaphor: reality (here, model output) flows like water through **channels**. Default channels — the model's high-probability generic outputs — are deep and low-resistance because they were carved by an enormous prior (pre-training data, “collective belief”). A desired non-default output is a **new channel**: initially high-resistance, it must be carved by sufficient *intention force* (guidance/steering/finetuning), after which it consolidates and becomes low-resistance itself. The whole theory is three equations; we lift them to model dynamics and show they unify the four sub-fields.

2. The Channel model

Let $\mathcal{M}$ be the model's output (or representation) manifold. A **channel** c is an attractor basin with depth s_c (how strongly trajectories fall into it) and resistance $R_c = 1/s_c$ (how hard it is to enter from outside). The AAMT equations (Core Premise §4.2):

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S_c      =  $\Sigma$  ( belief_strength · time_established · count )      # attractor depth
 $\Phi$       = intention_force / channel_resistance                      # flow into a channel
R_new(t) = R_0 · e{-t/ $\tau$ }                                           # carving a new channel
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In model terms:

- **s_c (depth)** = the prior mass on behavior c — for a default channel, the pre-training log-probability / frequency; for a representation, the strength of the learned feature. “Collective channels” (gravity, causality, default register) are deep; “individual channels” (a niche persona, a rare format) are shallow.
- **Φ (flow / steerability)** = how much output mass moves into a target channel given an applied force. `intention_force` is the magnitude of the inference-time push (guidance weight, steering-vector norm, prompt strength);
`channel_resistance` is the competing prior.
- **$R_{\text{new}}(t)$ (carving)** = the resistance of a *new* channel as it is reinforced, decaying exponentially with reinforcement time t and time constant τ set by coherence (Core Premise: $\tau \approx 7\text{--}14$ days at high coherence vs 90–180 at low — the model analog is high-vs-low learning-rate/quality regimes).

3. Three reductions

3.1 Steerability = the flow ratio (recovers guidance and activation steering)

A target behavior c^* competes with the default c_0 . The fraction of output mass captured by c^* under applied force f is monotone in

$$\Phi = f / R_{\{c^*\}} \quad \text{relative to} \quad f / R_{\{c_0\}}.$$

- **Classifier-free guidance.** With guidance weight w , the effective logit shift toward the conditioned channel is $w \cdot (\text{logit_cond} - \text{logit_uncond})$; identifying $f \propto w$ and $R_{\{c^*\}} \propto$ inverse prior of the conditioned behavior recovers the empirical fact that *rare* targets need *larger* w (high $R_{\{c^*\}}$) and that over-large w degrades quality (over-flow past the channel — the §4 “toxic positivity”/over-guidance failure). This is also the AAMT **30/70 rule**: $B = 1 - |\text{effort} - 0.3|$, an effort/flow optimum near 0.3 beyond which excess force raises resistance and lowers yield (Master Equation B -term; Foundations Paper III quadratic).
- **Activation steering.** A steering vector of norm f along a concept direction is exactly intention_force applied to that channel; CAA/RepE strength sweeps trace the Φ curve.

So “how steerable is behavior X” has a single quantity, Φ , with guidance and steering as instances differing only in *where* the force is applied (logits vs activations).

3.2 Mode/model collapse = carving the wrong channel (ties to WP-04)

Run the carving law on the *default* channel and you get collapse. When self-generated data re-enters training, the training flow re-deepens the channels the model already prefers: $S_{\{c_0\}}$ grows, $R_{\{c_0\}}$ falls as $e^{-t/\tau}$, and rare channels are starved. This is “model collapse” (Shumailov et al., 2024) expressed as runaway self-reinforcement of default channels. WP-04’s anti-feedback-collapse guard is exactly a bound on this: keeping the self-distillation fraction $f < 0.5$ (WP-04 §4.3, Rev. 2) keeps the *net* flow from compounding into the default channels — the contraction condition is the channel-carving stability condition. Channel Carving therefore gives WP-04’s guard a dynamical reading: prevent the wrong channels from carving themselves.

3.3 Consolidation/permanence = depth saturation (recovers 4τ)

A newly carved channel becomes “permanent” when its resistance has fallen enough that trajectories fall into it without applied force — i.e., $R_{\text{new}}(t) = R_0 e^{-t/\tau}$ has saturated. Taking the saturation horizon at $t = 4\tau$ (where $e^{-4} \approx 0.018$, ~98% carved) recovers the **4τ permanence** rule of Foundations Paper II and the 64-State stabilization theorem (Thm 5.1), and the collapse-risk function $P_{\text{collapse}}(t) = e^{-t/\tau}$

τ for $t < 4\tau$ is precisely $R_{\text{new}}(t)/R_0$. The same exponential also governs Hopfield attractor formation (WP-07) and new-channel carving — one τ -law across consolidation, permanence, and collapse.

Unifying statement. Steering (§3.1), collapse (§3.2), and permanence (§3.3) are the *same* attractor dynamics: Φ is the instantaneous flow into a channel;

$R_{\text{new}}(t) = R_0 e^{-t/\tau}$ is how a channel's resistance evolves under sustained flow; collapse is that law on default channels, permanence is its 4τ saturation on a desired channel.

4. Predictions and uses

1. **Steerability is predictable from prior mass.** A behavior's required guidance/steering force scales with its channel resistance $R_c \propto 1/\text{prior}(c)$; measuring prior log-prob should predict steering difficulty. (Testable: correlate steering-vector norm-to-effect with target rarity.)
 2. **A consolidation budget.** To make a finetuned behavior stick, reinforce for $\geq 4\tau$ of effective updates; below that, expect decay (the continual-learning forgetting curve is $e^{-t/\tau}$). This gives WP-04 a promotion criterion: *require a representation to survive 4τ cycles before promotion* (a Foundations Paper II testable prediction, now operational).
 3. **Collapse avoidance is a flow bound.** Keep net self-distillation flow into default channels under the contraction threshold ($f < 0.5$, WP-04); equivalently, keep fresh-data flow $\geq$ self-data flow.
 4. **Guidance optimum near 30/70.** Set inference-time force near the B -optimum (effort ≈ 0.3) rather than maximizing it; over-forcing raises effective resistance and lowers yield (quadratic, Paper III).
 5. **Carve, don't add.** Establishing a new channel needs *sustained* flow (carving), not a one-shot large push — the hysteresis result (Core Premise §6.5): graduated reinforcement reaches the target with less total energy than a single large jump.
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5. Relationship to prior work

Model/mode collapse (Shumailov et al., 2024; recursive-training analyses). Describe the *outcome* (distribution narrowing). Channel Carving gives the *mechanism* (default-channel self-deepening under the τ -law) and connects it to the same equation that governs desired consolidation. **Classifier-free guidance (Ho & Salimans, 2022).** An empirical knob; here it is the `intention_force` term of the flow ratio Φ , with the rarity-scaling and over-guidance degradation as consequences. **Activation steering / RepE (Zou et al., 2023; Rimskey et al., 2024).** The applied-force side of Φ in activation space. **Loss landscapes / mode connectivity / basins (Garipov et al., 2018; grokking analyses).** Provide the attractor-basin picture; Channel Carving adds the resistance/flow/ τ dynamics and the cross-phenomenon unification. **Hysteresis and consolidation (systems/continual learning).** The graduated-path advantage and $e^{\{-\tau/\tau\}}$ consolidation match the AAMT hysteresis and 4τ claims. **Key distinction.** No prior account unifies steerability, guidance, mode collapse, and consolidation/permanence under a single attractor-resistance law with one time constant τ , nor connects the consolidation horizon to a 4τ permanence rule.

6. Limitations

1. **Operationalizing `intention_force` and `R_c`.** We propose guidance weight / steering norm for force and inverse prior log-prob for resistance, but the exact functional forms need fitting per model.
 2. **τ is regime-dependent.** The carving time constant depends on learning rate, data quality, and coherence; the human-practice values (days) are illustrative, not numeric transfers.
 3. **Channel identification.** Treating a “behavior” as a single channel is an idealization; real behaviors are channel *networks*, and interference between them is future work (connects to the multi-concept orthogonalization of WP-06/WP-16).
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7. Conclusion

Channel Carving renders the AAMT Cardboard Channels theory as attractor dynamics and shows that steerability, guidance, mode collapse, and consolidation/permanence are one phenomenon: flow into attractor basins governed by $\Phi = \text{intention_force} / \text{channel_resistance}$ and resistance evolution $R_{\text{new}}(t) = R_0 e^{-t/\tau}$. Steering and guidance are the flow ratio; collapse is the law run on default channels (bounded by WP-04's $f < 0.5$ contraction condition); permanence is its 4τ saturation (Foundations Paper II; 64-State Thm 5.1). The framework yields concrete predictions — steering difficulty from prior mass, a 4τ consolidation budget for promotion, a 30/70 guidance optimum, and collapse-avoidance as a flow bound — and connects WP-04, WP-06/WP-16, and WP-07 through a single time constant.

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Conflict of Interest

The author is the founder and CEO of AsAManThinks Research.

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AAMT: Core Premise §4 (Cardboard Channels), §6 (hysteresis, thresholds); WP-04 (LACL collapse guard), WP-06/WP-16 (concept directions / sign-inversion), WP-07 (Hopfield watermark); Foundations Paper II (4τ permanence), Paper III (quadratic scaling); 64-State Quantum Oscillation System (Thm 5.1).